

Managing Projects to Achieve Results

Lesson 5





Study Information and Tips



This Study Text replicates the video content for the lesson



Supplementary knowledge can also be found in the information sheets and text boxes that accompany the videos, as well as in the resources in the Insights section



Activities to help you consolidate your knowledge are deliberately delivered through practice questions and exercises found in the Test and Apply sections on the VLE. These help you to further build on your knowledge and apply the learning in a way that supports your assessment



You can use this Study Text to test your understanding of the learning content at the end of each lesson. It may also be helpful to refer to it when preparing for your assessments



We recommend that you also produce your own notes in addition to the Study Text. Research has shown that handwritten notes can be very effective for learning retention



If you decide to print this Study Text, set the layout to print 4 or 6 slides per page to save on paper and ink



Managing Projects and Project Risks

How do we Identify Risks?

How do we Assess Risks and Develop Risk Registers?

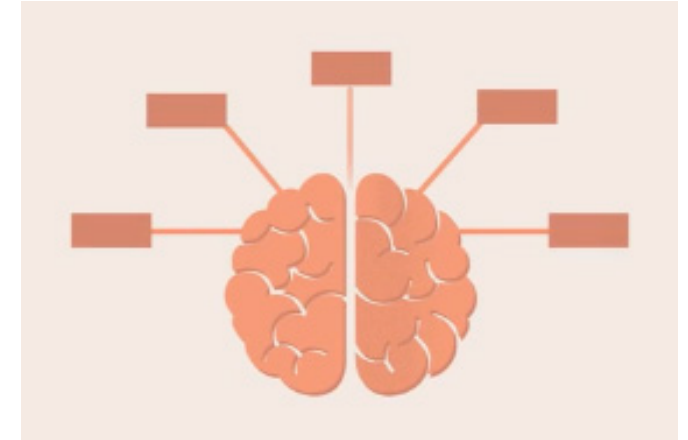
How do we Manage and Mitigate Project Risks?

Why Projects Succeed or Fail



Identifying Risks

Risk assessment is sometimes described as a transition from a world of many potential threats to a space where we have identified possible situations that could affect our organisation or project. There are many tools and techniques to help us with this. We usually draw on technical information, records and subjective evaluations to compile an initial list of risks. The most frequently used methods include:

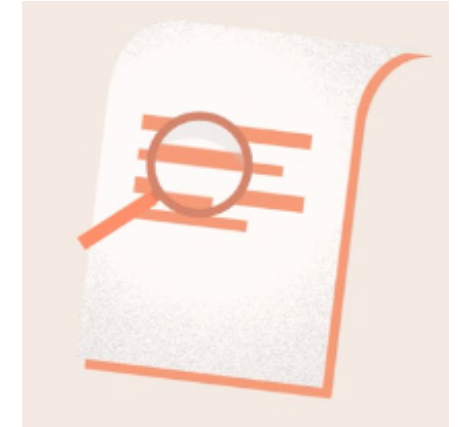


- Brainstorming. This is where people are encouraged to come up with their own risk ideas. You should try to include a range of people from across the organisation. In its book 'Managing Operations', the CMI recommends that these sessions have between six and fifteen participants, and last up to two hours. Risk ideas should be explained and written down during this meeting.



Identifying Risks (cont.)

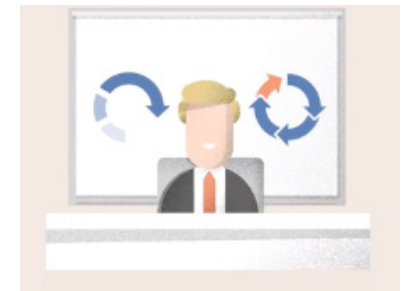
- There are also more formal, structured ways of conducting brainstorming sessions, including the HAZOP process, which stands for 'hazard' and 'operability'. Under the HAZOP model, participants are prompted to consider each step of an existing process and apply labels such as 'too early', 'too late', 'wrong action', 'wrong object', etc. These labels are used to identify potential hazards, and can help to keep discussions clear and focused.
- Historical analysis. This includes the examination of local or overseas experience with similar activities and projects. Some of this information may be publicly available; some may be published in paid-for journals. You might also be able to view post-project completion reports and audits for previous projects within your organisation.





Identifying Risks (cont.)

- Interviews and focus group discussions. These can be useful when your project has several important external stakeholders, or when the end result will be a consumer-facing product or service. They can provide rich qualitative data, as you can involve people in detailed discussions about problems, issues and potential risks.
- Scenario analyses. These are usually workshops, where key stakeholders propose 'scenarios' that are combinations of risks. They can be useful to draw attention to groupings of risk factors which may, individually, have been given little attention.
- Surveys and questionnaires. These have the advantage that you can collect more impartial data from a wider number and range of participants. Mostly, however, they are only 'one-way' – that is, you can't go back and pick up on interesting things that are said.





The Risk Register

Once you have attempted a few of these methods, you'll have a long list of potential risks. Some may seem very remote to you; others likely and important. Now your job is to enter them into the risk register.

The first columns are simply for tracking purposes. The third column is where we enter our identified risk.

The 'owner' field should contain the name of the most appropriate person to monitor and manage the risk. This person will also bear the consequences should the risk occur.

<i>Risks</i>			<i>Impact Analysis</i>		<i>Risk Management</i>		
<i>Risk Mitigation Strategy</i>	<i>Risk Category</i>	<i>Risk Identified</i>	<i>Severity</i>	<i>Likelihood</i>	<i>Quarterly Update</i>	<i>Further Action Required</i>	<i>Owner Field</i>



The Risk Register (cont.)

The 'risk category' fields allow you to assign a group to the identified risk. Examples include environmental, technical and operations. Categorising will help you to discern any obvious trends and allow you to devise strategies to mitigate them.

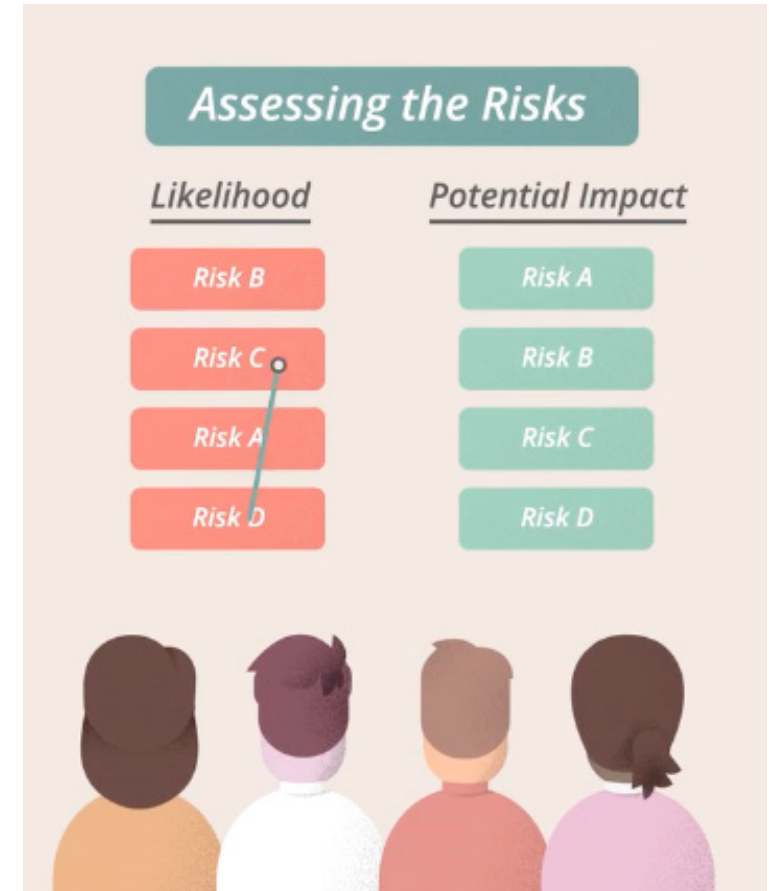




Assessing Risks

A fundamental part of this process involves assessing the likelihood and impact of risks, in the process assigning a low, medium or high risk tag to each.

In simple, smaller projects, assessing the risks may even be a part of the workshop in which you identify them. You might ask the participants to help you rank the risks both in terms of likelihood and potential impact. Be aware, however, that this process may well be subject to what is known as ‘confirmation bias’, people’s tendency to overemphasise ideas that support their point of view. So, for example, if you assemble a workshop full of engineers, you are likely to end up with a risk register that prioritises engineering risks far above financial and logistical problems.





Risk = Severity x Likelihood

A slightly more scientific approach is described by the Health and Safety Executive in guidance aimed at quarrying operations. It describes subjective risk assessment with a simple equation: Risk = Severity x Likelihood.

The process requires that you rank the severity of each risk you have identified on a scale of 1 to 3, with 1 being a major risk, and 3 being a slight risk. You then do the same for the likelihood of each risk becoming reality, again on a scale of 1 to 3, with 1 representing a high likelihood, and 3 a low likelihood.

Risk = Severity x Likelihood		
	<u>Severity</u>	<u>Likelihood</u>
Risk A	3	2
Risk B	2	1
Risk C	3	3
Risk D	1	1



Risk = Severity x Likelihood (cont.)

Completing the equation for each risk will then give you a risk value between 1 and 9 for each identified risk. Lower numbers will indicate severe and likely risks, which should be acted upon immediately.

The advantage of this approach over simply asking stakeholders to help you rank risk is that it allows you to evaluate them across two dimensions, and can be done quickly by multiple individuals. It also results in a single metric by which all risks can be compared.

	<u>Severity</u>		<u>Likelihood</u>		
Risk A	3	x	2	=	6
Risk B	2	x	1	=	2
Risk C	3	x	3	=	9
Risk D	1	x	1	=	1



Risk = Severity x Likelihood (cont.)

The disadvantages include the fact that it is still fairly subjective. For example, you or your stakeholders might identify an event such as a supplier going out of business, which, from a broader economic point of view is fairly common, and from a project point of view is obviously fairly serious. A risk like that might end up with a risk value of, say, 1 or 2, indicating an urgent priority.

Yet, this doesn't take account of how frequently in your project you might be exposed to such a risk. It might be that, in a construction project, you receive the bulk of your materials from one, large, reputable supplier, but once or twice you may be reliant on some smaller dealers. In this case, the risk value might be disproportionate to your project's actual 'exposure' to the risk.





Kinney-Wiruth Method

This was partly addressed by a semi-quantitative method developed by Kinney and Wiruth in the 1970s. They were working for the US Naval Weapons Centre, and wanted to help quantify the risks involved in testing new explosives.

They devised a new equation, which worked as follows. First, you assess the likelihood of the risk event occurring, on a scale ranging from 0.1 for 'virtually impossible', through to 10 for 'might well be expected'.

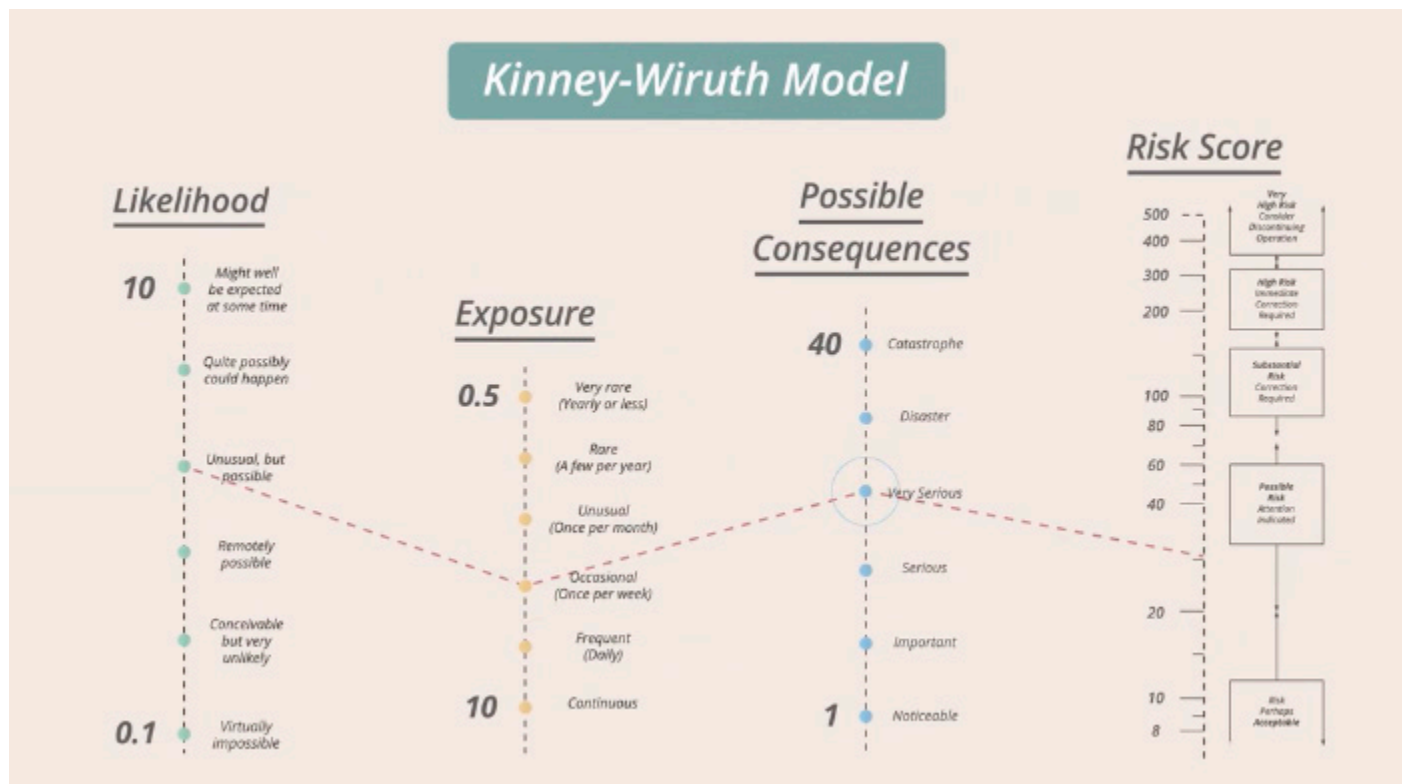
Then, you rank the likelihood of exposure to the risk, on a scale from 0.5 for a very rare, perhaps yearly, event, through to 10 for a continuous, everyday event. This is where the Kinney-Wiruth model becomes more powerful, because you have two measures of likelihood. You may also be able to feed in historic, or sector-specific, data here.

Finally, you rank the possible consequences for your event from 1 for 'noticeable', through to 40 for 'catastrophic'.



Kinney-Wiruth Method (cont.)

Then, you multiply all the factors together to get your risk score.





Kinney-Wiruth Method: Worked Example

Let's imagine that we have appointed a main contractor for doing the shop fit. We want to plan for the risk of this contractor going out of business. Using the Kinney-Wiruth scale, we might rank the likelihood of this occurring as 3: Unusual, but possible. This is quite high, and reflects the general churn of small businesses in the wider economy. However, let's suppose that we have access to a report from a trade body on the number of shopfitters that have filed for bankruptcy in the last five years. Reading this report has shown us that, for various reasons, this is actually quite unusual in this sector. So, we decide to give this risk an exposure factor of 1: 'a few per year'.

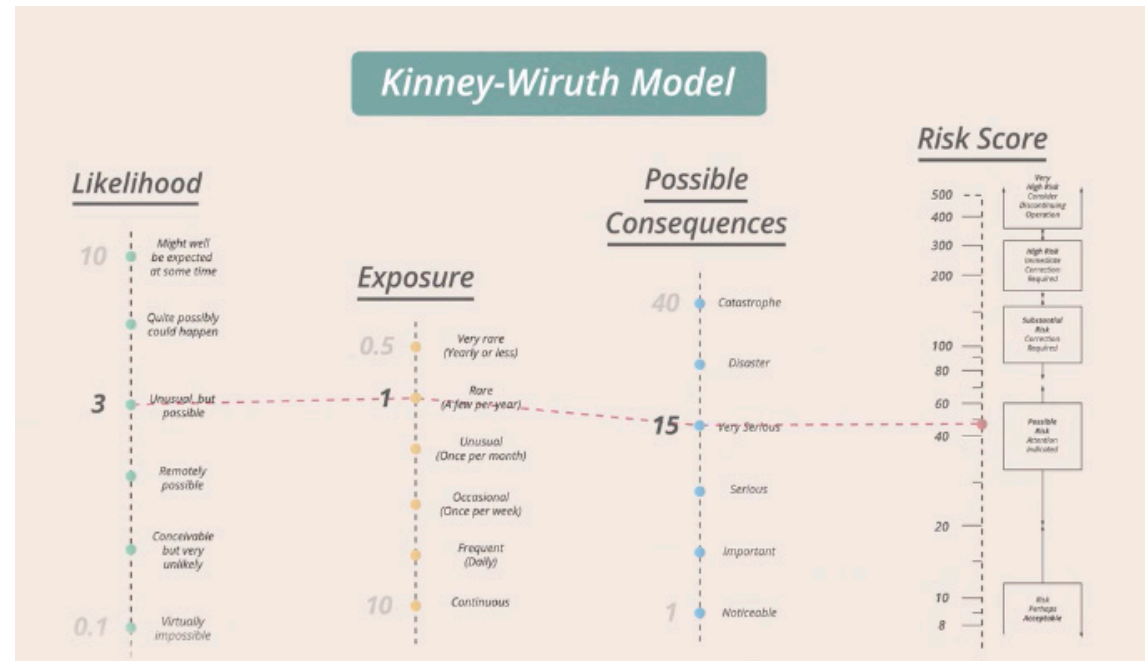


Finally, we rank the possible consequence of this risk as 15: very serious. We could find another shopfitter, but it would seriously delay our project, the opening of the new store, and our sales figures.



Kinney-Wiruth Method: Worked Example (cont.)

Multiplying these factors together gives $3 \times 1 \times 15 = 45$. Kinney and Wiruth give a scale that helps us evaluate the meaning of that score, which goes from less than 20 for an acceptable risk, through to 400+, which is a risk that should probably mean cancellation of the project. In our case, a risk of 45 falls into the 'possible risk: attention indicated' category. This is what we would expect – we should be aware of and plan for this risk, but equally it is a normal, anticipated danger for this kind of project.





Risk Assessment Techniques

So, you can see that even a semi-quantitative risk analysis has several advantages over a purely subjective approach. And there are a variety of steadily more sophisticated models (usually in the form of software packages) that may be required by your industry for accurately calculating risk.

Be aware, however, that more sophisticated does not automatically mean better results. In its Risk Assessment Techniques guidance, even the Industry Standards Organisation notes that:

“Full quantitative analysis may not always be possible or desirable due to insufficient information about the system or activity being analysed, lack of data, influence of human factors, etc. or because the effort of quantitative analysis is not warranted or required.”

Whatever method you choose to use, remember that all rely, to a certain extent, on human judgement, and that we must plan a mitigation strategy for every risk, even if the data suggest it will be small.



Risk Management and Mitigation

There are usually three options when it comes to mitigating risk:

1. You may choose to avoid the risk by removing its possible causes. This may or may not be possible.
2. You may take precautions to mitigate the impact of the risk.
3. You may accept the risk and its impact.



Risk Management and Mitigation (cont.)

The option you choose for each risk will depend on the risk score you calculated in your assessment process. For risks with lower scores, you may decide to accept the risk and its consequences. Risks with a high score – that is, a high likelihood of occurrence and with a large potential impact – should be dealt with first. It may be possible to avoid them, perhaps by redesigning part of the project, but usually it will be necessary to take precautions against them, in other words to mitigate them.

	<u>Severity</u>	<u>Likelihood</u>	<u>Risk = Severity x Likelihood</u>	
Risk A	3	2	$3 \times 2 = 6$	
Risk B	2	1	$2 \times 1 = 2$	
Risk C	3	3	$3 \times 3 = 9$	
Risk D	1	1	$1 \times 1 = 1$	



Risk Transfer and Risk Buffering

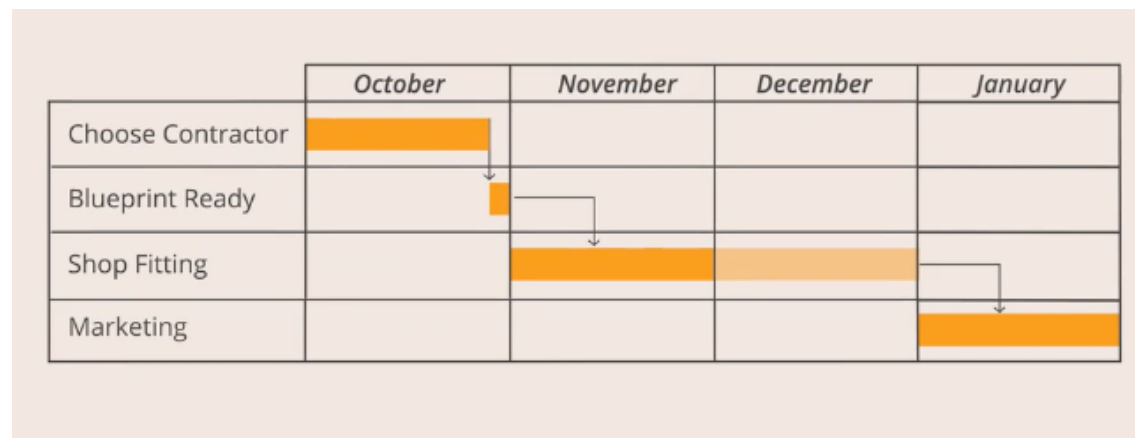
One option is known as risk transfer. Here, you shift some or all of the risk onto another party. In our example of the shop refit, we could attempt to share some of the risk of the venture with the contractor. We could do this by only paying a small amount to the contractor upfront, with progressively larger amounts of the balance to be paid as the project nears completion. The contractor may well quote a higher overall price in return for this risk-sharing arrangement, but that may be a price worth paying. Insurance is another, simpler, form of risk transfer.





Risk Transfer and Risk Buffering (cont.)

A second option is risk buffering. We know from our assessment that it is possible for a shopfitter to go bankrupt. If this were to happen, we estimate that it would take us about a month to appoint a replacement. Based on this, we could choose to give this phase of the project a one month buffer period. This helps to insure us against the risk, but will seriously reduce our post-refit revenues, because we won't be able to start marketing our new look and product-range for an extra month.





Risk Restructure

Finally, you might choose to restructure the risk. We could decide not to use a single shopfitting firm and appoint the architect, carpenters, electricians, and decorators individually. This would reduce our overall exposure to the risk of bankruptcy. But this approach would vastly increase the complexity of the project, and the possibility of knock-on delays. And complexity itself is a risk. As part of the restructuring, you would have to go back to the risk assessment process and evaluate which set of risks will have a smaller impact on the project were it to occur.



Don't view risk management and mitigation as a simple, one-shot process. There is no guarantee that you have identified all of the risks that may eventually affect your project.



Issues with Risk Management

In a study of a new iron-ore smelting project, De Meyer, Loch and Pich reported what happened as the project team ran into difficulties.

“Each time they had solved one problem, the next popped up just downstream in the chemical process. And worse, the problems predicted in the risk lists hardly occurred, while most problems were unexpected ones.”

This real world example teaches us two important things about risk management:

1. Although a risk register may make risks look like discrete, separate phenomena, they very often occur in combinations or sequences.
2. Many of the problems that you may actually encounter in a project are ones that you entirely failed to predict.



Three Essential Approaches to Risk

1. Instructionist. This is the standard ‘list and execute’ approach that we have already discussed and which still tends to form the mainstay of risk management.
2. Learning. A learning project is one that tries to stay as agile as possible, responding to unforeseen risks as they occur. The authors give the example of a mountaineering team, which, unable to reach the intended peak, decides to scale an alternative, smaller summit rather than return home empty-handed.
3. Selectionism. A selectionist approach is a sort of ‘survival of the fittest’ process for projects – you run a few different parallel approaches to solve a problem, and then, as soon as one clearly shows itself to be more promising (or less risky) than the others, you prioritise it. This is often an approach used by venture capitalists to manage their portfolio.

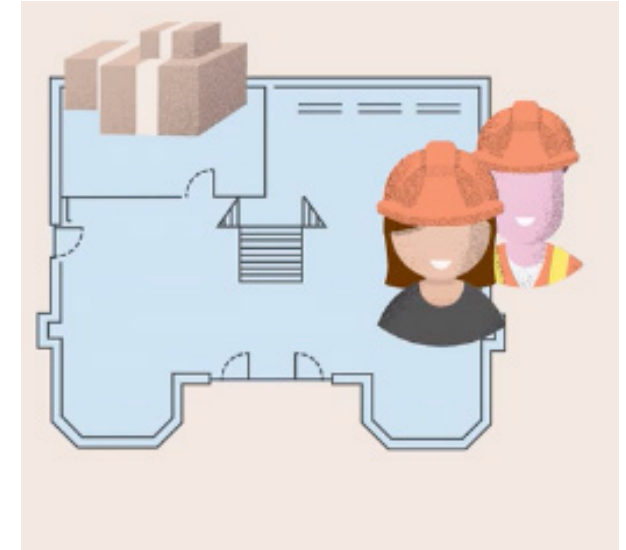
Selectionism may well only be available for larger-scale projects, or at an organisational level. But the learning project method has many applications.



Turn Risk into Opportunity

Let's return to our shop refurbishment project, and imagine that, halfway through the fit out, we hit a snag. We realise that the open-plan layout of the shop as originally designed will mean that we simply can't display enough stock, in turn putting strain on our stockroom and complicating our logistics. It could also affect our sales, as our customers like to browse.

Rather than ploughing ahead regardless just because we need to complete the project phase, a learning project should take the time to pause and reassess. Can the design be tweaked to include some extra shelving or display racks towards the rear of the store, without spoiling the aesthetic? Can we ask the carpenters to redesign an element of the stockroom to make it more space efficient? In other words, can we turn this unforeseen risk into an opportunity?





Why Projects Succeed or Fail

We've looked at the many ways that you can plan, budget, manage and monitor your project in order to make it a success. The reality is, however, that most projects will fail.

In business, some people regard focusing on failure as an indicator of a negative attitude. But it is only by learning the lessons offered by these failures that we can ensure those mistakes are not repeated. We're going to look briefly at what happens when a project goes wrong, and how you can avoid some of those situations.



Why Projects Succeed or Fail (cont.)

Common problems in project management include:

- Not establishing a valid business case for the project
- Giving insufficient attention to quality
- Outcomes that were not clearly defined
- A lack of communication with stakeholders
- No clear definition of project management roles and responsibilities
- Poor estimation of duration and costs
- Inadequate planning and coordination of resources
- Insufficient performance measurements and lack of control over progress
- A manager being so obsessed with a need to succeed that project warning signs were just filed away and ignored



Examples of Failure

You will no doubt be aware of some high-profile examples of projects that have failed with massive cost overruns, late completion dates or lacking the intended quality or functionality. Examples include:

- Cost – The NHS patient record system, was initially expected to cost £6.4 bn, but its final cost is in excess of £10bn
- Time – nuclear power station construction frequently runs late – in 2014, of the 66 nuclear reactors ‘under construction’ worldwide, 49 were running behind schedule
- Quality – the initial images from the Hubble space telescope were only slightly better than those from Earth-based telescopes, because the most precisely ground mirror in history had actually been ground into the wrong shape

For a project to be successful, it must be completed on time, on budget, and to the desired level of quality.





Evaluating Your Project Plan

A useful approach to evaluating project plans was devised by Elbeik and Thomas in 1998. In order of priority, their indicators of success are:

1. Clearly defined objectives
2. Good planning and control method
3. Good quality of project manager
4. Good management support
5. Enough time and resources
6. Commitment by all
7. High user involvement
8. Good communications
9. Good project organisation and structure
10. Being able to stop a project





Evaluating Your Project Plan (cont.)

You might be surprised that the ‘quality of the project manager’ is only number three on the list, and that ‘clearly defined objectives’ and ‘good planning and control’ come first. While those could be the result of the actions of a good project manager, once they are properly set in place, the project may then be managed with a light touch.



Recap

In this lesson, we've looked at:

- How to identify project risks
- Assessing their likelihood and impact
- Planning mitigation strategies
- Why projects succeed or fail
- Evaluating your project plan